

Spherical Bodies: World Shape Lab

I can collect fair comparison observations and combine them with real space evidence to describe the Sun, Earth and Moon as approximately spherical.

Same viewpoints

use evidence

Six observations

use evidence

Evidenced conclusion

use evidence

[Open the original interactive lesson](#)

What success looks like

- Observe a disc and sphere from the same three viewpoints.
- Record similarities and differences without changing the claim to fit a guess.
- Combine model and real space evidence in a conclusion with one honest limitation.

Retrieve and reconnect

- Last lesson: why was one circular view insufficient?
- Last lesson: what happened to the disc outline as it turned?
- Last week: how did Earth's curved surface affect light and darkness?
- Lead-in: what must stay the same for a fair shape comparison?

Which result would most strongly challenge a flat-disc model? Commit before seeing the method.

Think first. Point, say, sign or write your choice. Give one reason.

- A. The body looks circular from the front.
- B. The outline stays round from front, quarter and edge directions.
- C. The object can move across a table.

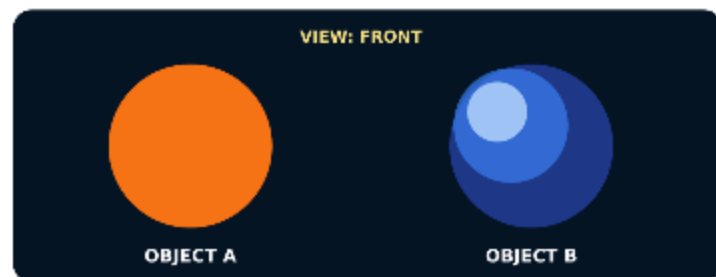
Look closely · what is the evidence?

Observation 1: both objects look round from the front.

Observation 2: after the same quarter turn, one looks oval and one stays round.

Observation 3: from the edge direction, one is a thin line and one stays round.

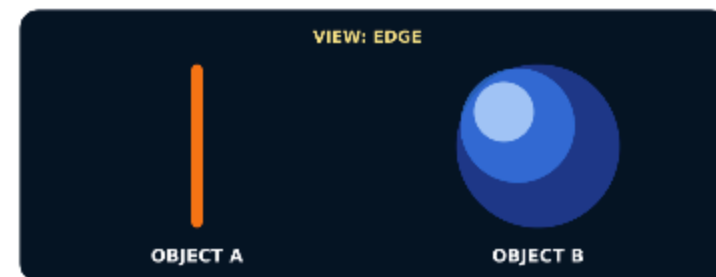
Look closely · what is the evidence?



Both objects round from the front



One oval, one round after the same turn



One a line, one round from the edge

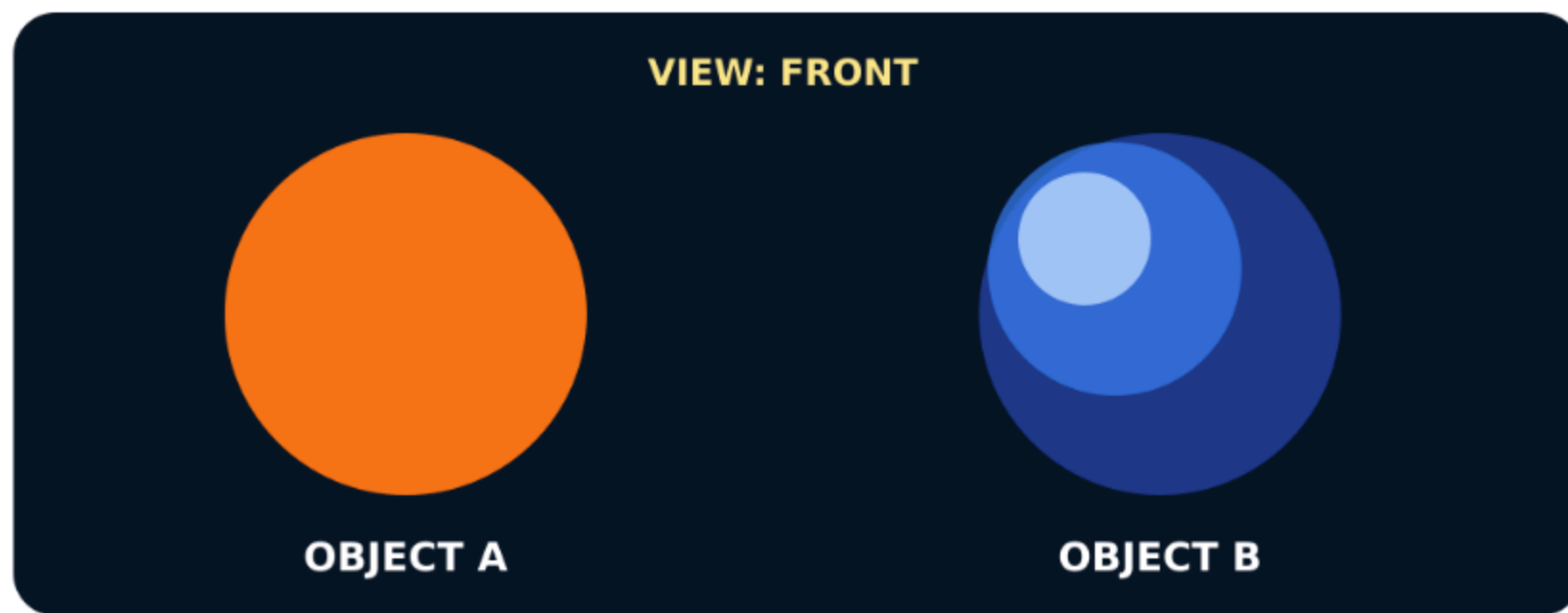
Words we will use

- observation — what is noticed
- comparison — same test for both
- conclusion — what the evidence means

Watch the model

- Place the disc and sphere at the same marked distance. Observe both from the front without naming the expected answer.
- Turn both through the same quarter turn, then to the edge direction. Record only the outline seen each time.
- Check the six observations before concluding. A fair comparison uses the same viewpoints for both objects.

Watch the model



A disc and a sphere seen from the chosen viewpoint

Which method gives the fairest shape comparison?

- A. View the disc close-up and the sphere from across the room.
 - B. Use the same distance and the same three viewpoints for both.
 - C. View only the front of whichever object looks roundest.
- Commit to an answer. Show the evidence you used. Listen, then revise if needed.

Build the explanation

- Turn the six model observations into a comparison: the disc outline changes; the sphere's outline stays round.
- Add real observations: spacecraft views keep showing round whole-body outlines, Earth's eclipse shadow is curved, and the curved day-night edge moves across Earth and the Moon.
- Make the careful claim: gravity makes large bodies nearly round; rotation can flatten them slightly, and surface features add small bumps. The Sun, Earth and Moon are approximately spherical.

Find and repair the misconception

Conclusion glitch: the Sun, Earth and Moon are perfectly smooth, perfect spheres.

A. Keep it: every spherical body must be perfectly smooth.

B. Repair it: they are approximately spherical, with real surface features and small departures from perfection.

C. Repair it: surface features make all three bodies flat.

Six-observation world shape lab

Sort the model results and real space observations, then freeze enough evidence for a careful conclusion.

- Choose one recorded observation.
- Commit it to disc, sphere or real space evidence.
- Repair using the named view or sky observation.
- Freeze model and real evidence, then conclude.

Use the numbered cards in your pupil pack. Take turns to choose, justify and check. You may direct a partner to move a card.

Choose a group · defend your placement

- Disc record
- Sphere model record
- Real space record

Use one paired model view and one real space observation in your conclusion.

Your independent mission

Complete the fair three-view comparison and present an evidence-led conclusion about Solar System body shape.

Record: Six model observations, one named real space observation, an approximately spherical conclusion and one model limitation.

Choose your support and challenge

- Supported: Complete a six-row picture table and choose the conclusion supported.
- Standard: Record all six model observations, add one real space observation and justify the approximately spherical description.
- Stretch: Evaluate the evidence, identify an alternative explanation from one view and state why combined views are stronger.

Show what you know

Cite one paired model view and one real space observation to justify approximately spherical.

Point to, read or explain the evidence you made. Choose one next step after the adult has listened.